

08/10/26

Activity-4 Violin Skewness

Dudekula Mohammad Ilyas - 192325017 Roll no: 13

Part-A: Identify skewness & mean vs median

Right Skewed: Tail extends to right, $\text{mean} > \text{median}$

Left Skewed: Tail extends to left, $\text{mean} < \text{median}$

Symmetric: both sides are approximately balanced, $\text{mean} \approx \text{median}$

Chart 1

Violin	Skewness	mean vs median
Station 2	Left-skewed	$\text{mean} < \text{median}$
Station 3	Right skewed	$\text{Mean} > \text{median}$
Station 5	Symmetric	$\text{mean} \approx \text{median}$

Chart 2

Violin	Skewness	mean vs median
Class C	Right skewed	$\text{mean} > \text{median}$
Class A	Left-skewed	$\text{mean} < \text{median}$
Class E	Symmetric	$\text{mean} \approx \text{median}$

Chart 3

Violin	Skewness	mean vs Median
Class C	Right Skewed	$\text{Mean} > \text{median}$
Class A	Left Skewed	$\text{Mean} < \text{median}$
Class E	Symmetric	$\text{mean} \approx \text{median}$

Chart 4

Violin	Skewness	mean vs median
Sewer 2	Symmetric	$\text{mean} \approx \text{median}$
Sewer 1	Left Skewed	$\text{mean} < \text{median}$
Sewer 4	Right skewed	$\text{mean} > \text{median}$

Part-B

B1: The tail extends towards lower no. of sleep hours, so it is left-skewed. The median is a better measure of typical sleep because it is less affected by unusually low values.

B2: `ggplot(df, aes(X=group, y=value)) +
geom_violin() +
stat_summary(fun = median, geom = 'point', color = 'white')`